



American International University-Bangladesh (AIUB)
Faculty of Engineering
Department of CSE, EEE, and CoE

EEE4103 MICROPROCESSOR AND EMBEDDED SYSTEM
COURSE CAPSTONE PROJECT PROPOSAL FORM

SEMESTER: SPRING 2025-26

PROJECT TITLE: (must be an embedded system designed using Arduino/STM32/Raspberry Pi with other necessary sensors, actuators, components, etc. Both hardware implementation and simulation must be shown by the end of the semester. However, the proposal form should be submitted within the next two weeks from now on) [2 MARKS]

SURVEY: (to develop a process for complex engineering problems considering cultural and societal factors (use **Google or Microsoft Forms** to design questions, show the outcomes in the form of a pie/bar chart, show the links of questions, **minimum 10 questions, and 40 respondents**), present survey link) [7 MARKS]

AIMS AND OBJECTIVES OF THE PROJECT: (there must be a minimum of three aims and objectives of your work using measurable **action verbs** of Bloom's Taxonomy, for example, design, simulate, implement, analyze, etc.) [4 MARKS]

LITERATURE REVIEW: (must be related to embedded system design with other necessary citations from the most recently published papers. At least 5 most recent publications must be collected, studied, identified the problems, and then discussed in this section) [5 MARKS]

EXPERIMENTAL BLOCK DIAGRAM: (demonstrate how this project is interconnected with various components and how various signals flow among them using text boxes and arrows) [5 MARKS]

POSSIBLE OUTCOMES OF THE PROJECT: (describe how this project can help society and improve the culture of society) [4 MARKS]

PROJECT TIMELINE (GANTT CHART): describe how this project will progress with time by breaking the whole task into several components concerning some specific time using a horizontal or vertical bar graph) [5 MARKS]

REFERENCES: (only published paper-based references are allowed with the IEEE format, **don't use YouTube links, Wikipedia, or any random websites for references**; however, datasheets, databases, can be used as references with the appropriate formats, links, and access dates): [3 MARKS]

Sample for the conference paper:

[1] M. H. Bhuyan and Q. D. M. Khosru, "Linear Asymmetric Pocket Profile Based Pinch Off Voltage Model for Nano Scale n-MOSFET," Proceedings of the IEEE sponsored International Conference on Electrical, Computer and Communication Engineering (ICECCE2017), organized by the Chittagong University of Engineering and Technology (CUET), Cox's Bazar, Bangladesh, 16-18 February 2017, pp. 28-32.

Sample for the journal paper:

[1] M. H. Bhuyan and Q. D. M. Khosru, "Effects of Temperature on Reverse Short Channel Effect in Pocket Implanted Sub-100 nm n-MOSFET," Journal of Materials Science and Engineering, USA, 1934-8959, vol. 4, no 7, July 2010, pp. 18-23, DOI:10.17265/2161-6213/2010.07.004.

Course Name:	Microprocessor and Embedded Systems Lab	Course Code:	EEE 4103
Semester:	Spring 2025-26	Sec:	U
Faculty Member:	DEBRAJ DAS		

Capstone Project Title:	Secure Arduino-Based Electronic Voting Machine with Face Recognition		
Project Group No.	04		
Sl #	Student ID #	Student Name	Obtained Marks
1.	23-54966-3	Fabiha Lamisa Tridha	

Assessment Rubrics:

KPIs	Excellent [2]	Proficient [1.5]	Good [1]	Acceptable [0.5]	Unacceptable [0]	No Response [0]	Secured Marks
Project Title	The title reflects an issue related to complex engineering problems, showing targets and methods with possible outcomes.	The title reflects an issue related to complex engineering problems, showing targets and methods, but some missing issues.	The title reflects an issue related to the course capstone project, but there may be some missing issues.	The title reflects an issue related to the course capstone project, but it is not complete or specific.	The title does not reflect any issues related to the course capstone project.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						Total Marks (2)	

KPIs	Excellent [6-7]	Proficient [4-5]	Good [3]	Acceptable [2]	Unacceptable [1]	No Response [0]	Secured Marks
Survey	The survey developed as a process for complex engineering problems, considering cultural and societal factors, has superior variables, targets, measures, and the implementation process is clear and challenging for future project implementation, with several possible outcomes having good impacts.	The survey, developed as a process for complex engineering problems, considering cultural and societal factors, has good variables, targets, measures, and the implementation process is clear and challenging for future project implementation, with some possible outcomes and little impact.	The survey developed as a process for complex engineering problems, considering cultural and societal factors, has moderate variables, targets, measures, and The implementation process is clear and challenging for future project implementation, with a few possible outcomes and impacts.	The survey, developed as a process for complex engineering problems, considering cultural and societal factors, has good variables, targets, measures, and the implementation process is somewhat clear for future project implementation, with very few possible outcomes and little impact.	The survey developed as a process for complex engineering problems, considering cultural and societal factors, has poor variables, targets, measures, and the implementation process is very unclear for future project implementation, with few possible outcomes but no impacts.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						Total Marks (7)	

KPIs	Excellent [4]	Proficient [3]	Good [2]	Acceptable [1]	Unacceptable [0.5]	No Response [0]	Secured Marks
Aims and Objectives	Aims and objectives are written to solve complex engineering problems, considering cultural and societal factors with specific targets, measurements, and implementation processes that are clear and challenging, and have several possible outcomes having very good impacts.	Aims and objectives are written to solve complex engineering problems, considering cultural and societal factors with general targets, measurement, and implementation processes that are not clear and challenging, and have some possible outcomes that have a good impact.	Aims and objectives are written to solve complex engineering problems considering a few cultural and societal factors with narrow targets; measurement, and implementation processes are clear and challenging and have a few possible outcomes, having some impacts.	Aims and objectives are written to solve complex engineering problems considering cultural or societal factors with a very specific target; measurement and implementation processes are not clear or challenging, and have little possible outcome, having no impact.	Aims and objectives are written to solve complex engineering problems, but do not consider cultural and societal factors with any targets; measurement and implementation processes are not clear and challenging, and no possible outcomes that have an impact.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						Total Marks (4)	

KPIs	Excellent [5]	Proficient [4]	Good [3]	Acceptable [2]	Unacceptable [1]	No Response [0]	Secured Marks
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Literature Review	Specific formats are maintained to review and cite the literature with recent publications. Identified and analyzed the problem correctly.	Specific formats are maintained to review and cite the literature with recent publications. Identified and analyzed the problem correctly, but not all issues were addressed with relevant or intended work.	Specific formats are maintained to review and cite the literature with recent and past publications. Identified and analyzed the problem correctly, but not all issues were addressed with relevant or intended work.	Specific formats are maintained to review and cite the literature with recent and past publications. Identified but could not analyze all the problems correctly, and all issues were not addressed with relevant or intended work.	No specific formats are maintained to review and cite the literature with recent publications. Could not identify and analyze all the problems correctly, and all issues are not addressed with relevant or intended work at all.	No Response at all/ copied from others/ identical submissions with gross errors/ image file printed	
Comments						Total Marks (5)	

KPIs	Excellent [5]	Proficient [4]	Good [3]	Acceptable [2]	Unacceptable [1]	No Response [0]	Secured Marks
Experimental Block Diagram	The block diagram is drawn to show the connections of all the possible components or subsystems to show their interdependence with all possible flows of signals from inputs to outputs.	The block diagram is drawn to show the connections of all of the possible components or sub-systems to show their interdependence, with a few missing flows of signals from inputs to outputs.	The block diagram is drawn to show the connections of most of the possible components or sub-systems to show their interdependence, with a few missing flows of signals from inputs to outputs.	The block diagram is drawn to show the connections of a few possible components or sub-systems to show their interdependence, with some missing flow of signals from inputs to outputs.	The block diagram is not drawn to show the connections of all possible components or sub-systems, to show their interdependence and flow of signals from inputs to outputs.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						Total Marks (5)	

KPIs	Excellent [4]	Proficient [3]	Good [2]	Acceptable [1]	Unacceptable [0.5]	No Response [0]	Secured Marks
Possible Outcomes	Outcomes are written to achieve complex engineering problems' solutions, considering cultural and societal factors and showing measurement and implementation processes to attain the outcomes with all possible impacts.	Outcomes are written to achieve complex engineering problems' solutions, considering cultural and societal factors and showing measurement and implementation processes to attain the outcomes with some impacts.	Outcomes are written to achieve complex engineering problems' solutions, considering cultural and societal factors, and do not show measurement and implementation processes to attain the outcomes without showing any impacts.	Outcomes are written to achieve complex engineering problems' solutions, but do not consider cultural and societal factors, and do not show measurement and implementation processes to attain the outcomes without showing any impacts.	Outcomes are not written to achieve complex engineering problems' solutions, do not consider cultural and societal factors, and do not show measurement and implementation processes to attain the outcomes without showing any impacts.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						Total Marks (4)	

KPIs	Excellent [5]	Proficient [4]	Good [3]	Acceptable [2]	Unacceptable [1]	No Response [0]	Secured Marks
Gantt Chart	Specific formats are maintained to draw the Gantt chart, and there is an order of workflow with all work to be done.	Specific formats are maintained to draw the Gantt chart, and there is an order of workflow with a few tasks missing.	Specific formats are maintained to draw the Gantt chart, and there is an order of workflow with some work missing.	No specific formats are maintained to draw the Gantt chart, and there is little order to the workflow, with some work missing.	No specific formats are maintained to draw the Gantt chart, and there is no order of workflow, with the most important tasks missing.	No Response at all/ copied from others/ identical submissions with gross errors/ image file printed	
Comments						Total Marks (5)	

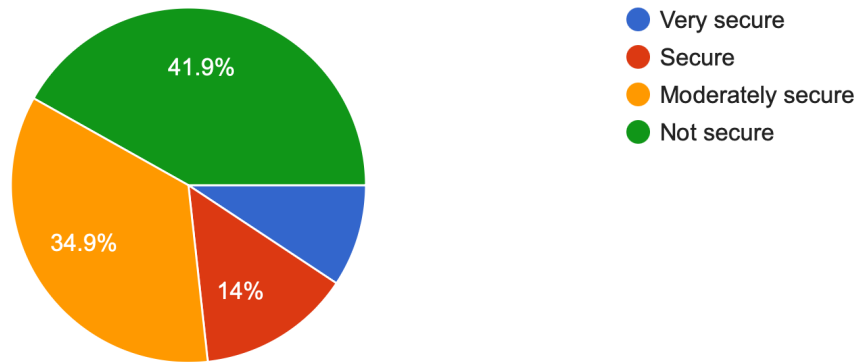
KPIs	Excellent [3]	Proficient [2.5]	Good [2]	Acceptable [1]	Unacceptable [0.5]	No Response [0]	Secured Marks
References	Specific formats are maintained to write references, and all are recently published journal and conference papers having no missing information.	Specific formats are maintained to write the references, and all are journal and conference papers, but some old papers have missing information.	No specific formats are maintained to write the references, and many are internet sources with several missing information and very old references.	No specific formats are maintained to write the references, and most of them are internet sources with missing information.	No specific formats are maintained to write the references, and all are internet sources with missing information.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						Total Marks (3)	

Title: Arduino-Based Electronic Voting Machine Using Face Recognition

Survey: <https://forms.gle/15fDfnSCDgqqnwDQ7>

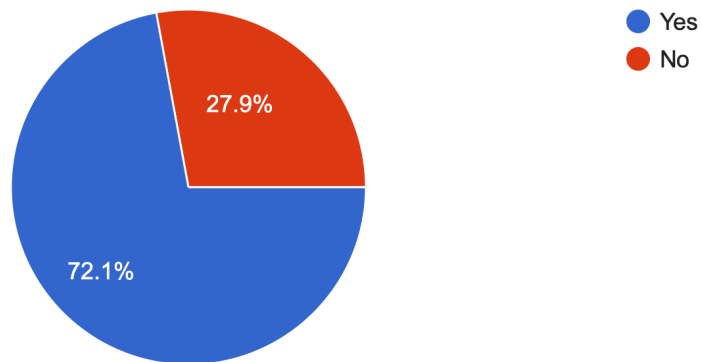
How secure do you think the current voting system in Bangladesh is?

43 responses



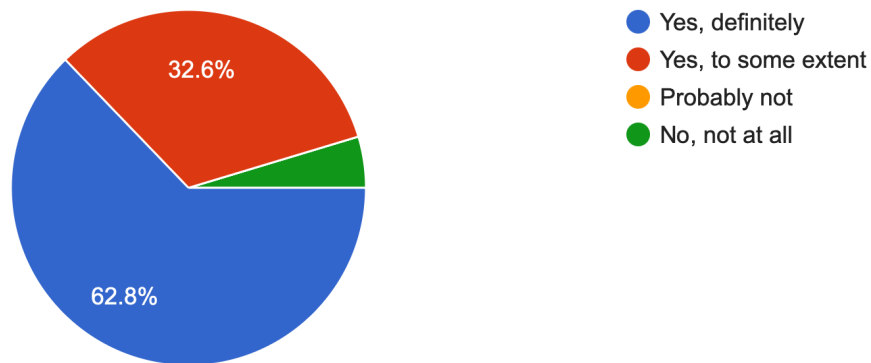
Do you trust electronic voting machines for national elections?

43 responses



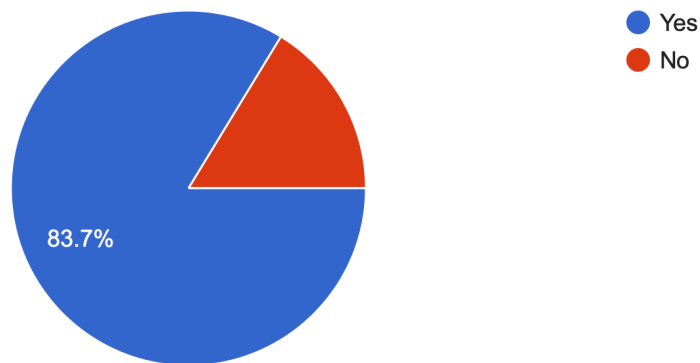
Do you think face recognition technology can improve voting security?

43 responses



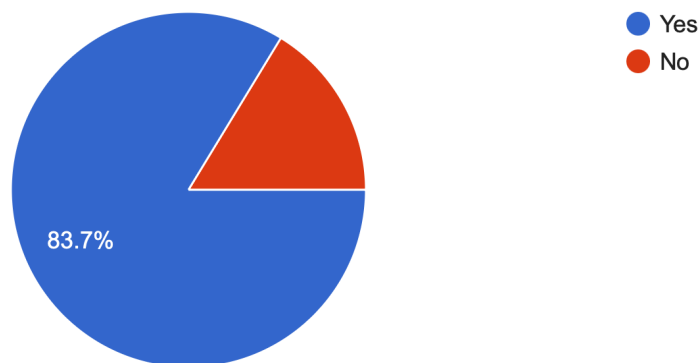
Do you think electronic voting systems can prevent duplicate voting?

43 responses



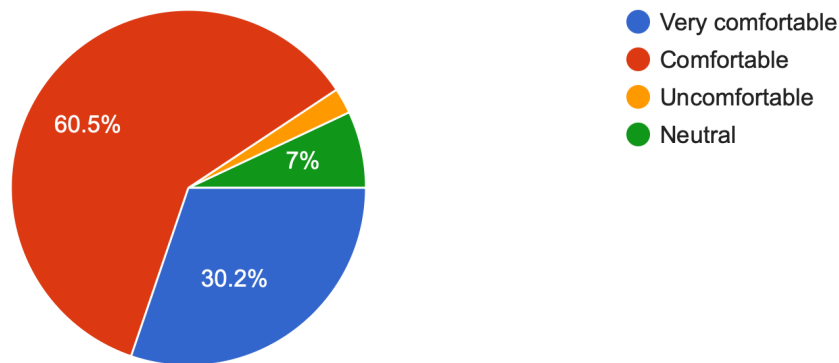
Do you think electronic voting systems can prevent duplicate voting?

43 responses



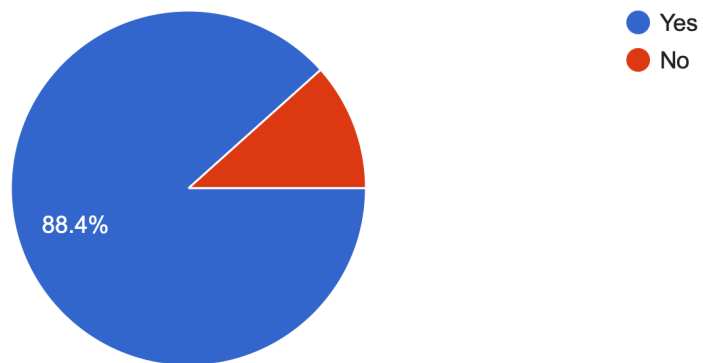
How comfortable would you feel using face recognition during voting?

43 responses



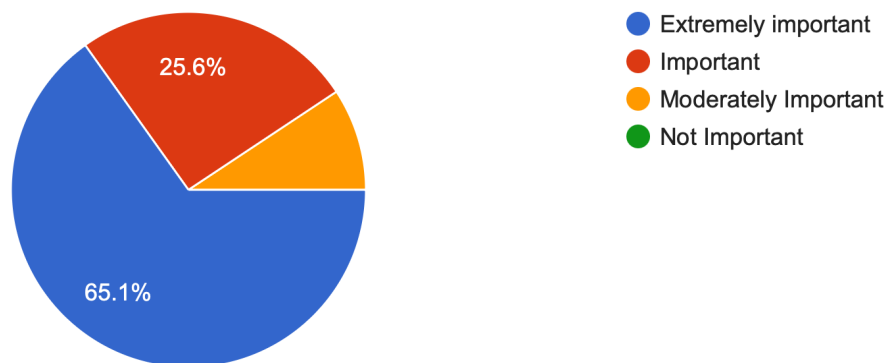
Do you think technology can reduce election fraud?

43 responses



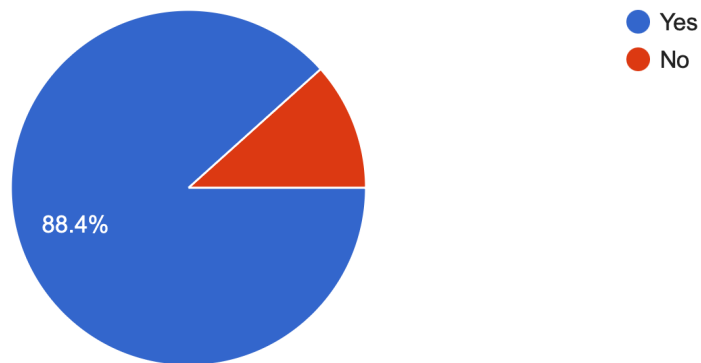
How important is voter authentication before casting a vote?

43 responses



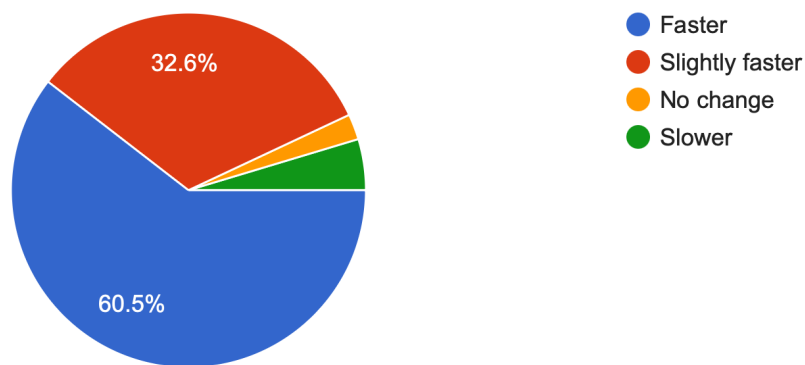
Do you support implementing electronic voting systems in future elections?

43 responses



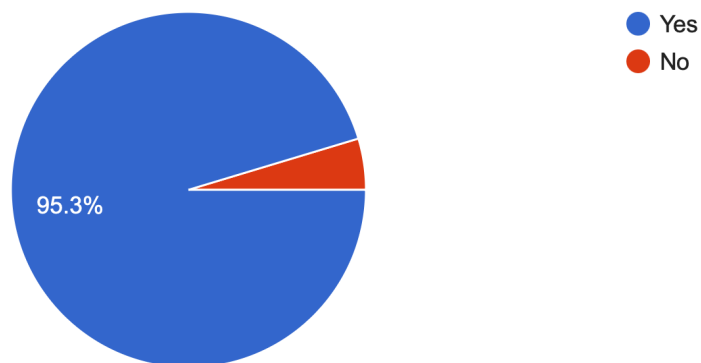
How much faster do you think automated voting systems can make elections?

43 responses



Should biometric technologies like face recognition be used in elections?

43 responses



Objectives:

1. To design a secure electronic voting machine using Arduino and face recognition technology.
2. To implement a voter authentication system using real-time face recognition before allowing voting access.
3. To simulate and integrate communication between the computer-based face recognition system and Arduino hardware.
4. To analyze the effectiveness of face recognition in preventing duplicate voting and unauthorized access.

LITERATURE REVIEW

Paper 1: Smart Voting System Using Deep Learning Techniques and Facial Authentication (2022)

- Proposed an electronic voting system that uses deep learning–based face recognition for voter authentication before allowing vote casting.
- The system captures the voter’s facial image and compares it with stored facial data to verify identity.
- **Problem:** Recognition accuracy may decrease under different lighting conditions or when facial expressions vary.
- **Contribution to this project:** Demonstrates the feasibility of integrating facial authentication with voting systems to prevent voter impersonation and improve election security.

Paper 2: Face Recognition Based Electronic Voting System (2024)

- Developed an electronic voting framework that uses facial recognition to verify voter identity through a webcam-based system.
- The system extracts facial features and compares them with a stored database before allowing a user to vote.
- **Problem:** Large voter databases can increase recognition time and reduce system performance.
- **Contribution:** Shows how face recognition technology can be used to ensure that only registered voters participate in elections.

Paper 3: Online Voting System Using Face Recognition and Fraud Detection (2025)

- Proposed a voting platform where voters are authenticated through facial recognition combined with fraud detection algorithms.
- The system detects suspicious voting attempts and prevents multiple voting using database validation.
- **Problem:** Biometric systems require strong protection against spoofing attacks such as using photos or videos.
- **Contribution:** Highlights the importance of implementing secure authentication and fraud detection mechanisms in voting systems.

Paper 4: Biometric Voting Solution Integrating Face Recognition with Embedded Systems (2025)

- Designed a biometric voting system where facial recognition is integrated with embedded hardware for real-time voter authentication.
- The system achieved high recognition accuracy and ensured the “one person–one vote” rule.
- **Problem:** Requires optimized processing for real-time performance in embedded systems.
- **Contribution:** Demonstrates that face recognition can be successfully integrated with embedded platforms for secure offline elections.

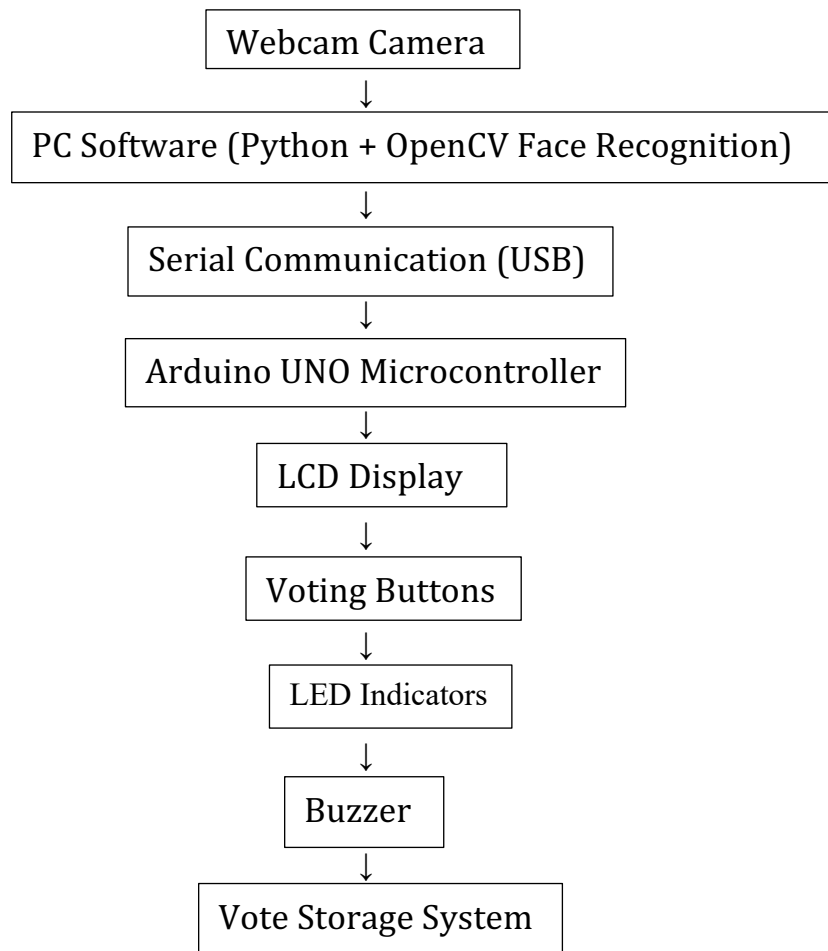
Paper 5: Blockchain and Face Recognition Based E-Voting System (2025)

- Proposed a decentralized electronic voting system combining blockchain technology with facial recognition for voter authentication.
- The system ensures vote integrity and prevents data manipulation using secure distributed storage.
- **Problem:** Blockchain integration increases computational complexity and system cost.
- **Contribution:** Provides insights into improving election transparency and security through biometric authentication.

Identified Gaps

- Many existing electronic voting systems rely only on software-based face recognition without embedded hardware integration.
- Several systems require high computational resources or cloud infrastructure, making them difficult to implement in low-cost environments.
- Some studies do not provide real-time communication between authentication software and hardware voting interfaces.
- Many prototypes lack simple, low-cost implementations suitable for educational institutions or small-scale elections.

EXPERIMENTAL BLOCK DIAGRAM



Signal Flow Description

1. The webcam captures the voter's face and sends the image to the computer.
2. The Python program processes the image using face recognition algorithms.
3. The system compares the captured face with stored voter data.
4. If the voter is authenticated successfully, the computer sends a signal to the Arduino through serial communication.
5. Arduino activates the voting interface and displays instructions on the LCD screen.
6. The voter selects a candidate using the voting buttons.
7. The vote is recorded, and the buzzer confirms successful vote submission.

POSSIBLE OUTCOMES OF THE PROJECT

This project will demonstrate a secure prototype of an electronic voting machine integrated with face recognition technology. The system will allow only authenticated voters to cast votes, reducing the risk of voter impersonation and duplicate voting.

By integrating computer vision and embedded systems, the project can contribute to the development of smarter election systems that are more transparent, efficient, and reliable. Such systems could improve public trust in election processes and encourage the adoption of advanced technology in future voting systems.

PROJECT TIMELINE (GANTT CHART):

Task	Feb	Mar	Apr	May
Literature Review				
Survey Design & Data Collection				
System Design & Component Selection				
Face Recognition Algorithm Development				
Hardware Setup (Camera, Microcontroller, Sensors)				
System Integration				
Testing and Debugging				
Performance Evaluation				
Documentation and Final Report				

References

- [1] S. S. Raut, P. M. Jawandhiya, and S. R. Suralkar, “Smart Voting System Using Deep Learning Techniques and Facial Authentication,” *International Journal of Engineering Research & Technology (IJERT)*, vol. 11, no. 5, pp. 559–563, May 2022.
- [2] R. Patel, A. Shah, and D. Mehta, “Face Recognition Based Electronic Voting System,” *International Journal of Information Technology and Computer Engineering (IJITCE)*, vol. 6, no. 2, pp. 15–20, 2024.
- [3] A. Kumar, S. Verma, and P. Singh, “Online Voting System Using Face Recognition and Fraud Detection,” *International Journal of Engineering Research & Technology (IJERT)*, vol. 14, no. 3, pp. 421–426, 2025.
- [4] M. A. Rahman, T. Islam, and S. Ahmed, “Biometric Voting Solution Integrating Face Recognition with Embedded Systems,” *Journal of Engineering Research and Reports*, vol. 25, no. 4, pp. 45–52, 2025.
- [5] R. Sharma and K. Gupta, “A Blockchain and Face Recognition Based E-Voting System,” *International Journal for Research in Applied Science and Engineering Technology (IJRASET)*, vol. 13, no. 1, pp. 120–125, 2025.
- [6] G. Bradski, “The OpenCV Library,” *Dr. Dobb’s Journal of Software Tools*, vol. 25, no. 11, pp. 120–126, Nov. 2000.
- [7] Arduino, “Arduino UNO Technical Specifications,” Documentation for Arduino Uno, Available: <https://www.arduino.cc>, Accessed: Mar. 2026.